# smARTvalve / smARTsynth MIDI

**NOTES**

- Control 60 - 64 are free to respect standard Sustain Pedal message.
- Control 74 is not used to respect default MPE Slide/Aftertouch CC

**Headroom** selects the maximum of wind pressure reachable with Pressure (40) . It's max is %200 if wind pressure (over pressure) and its middle point its %100.

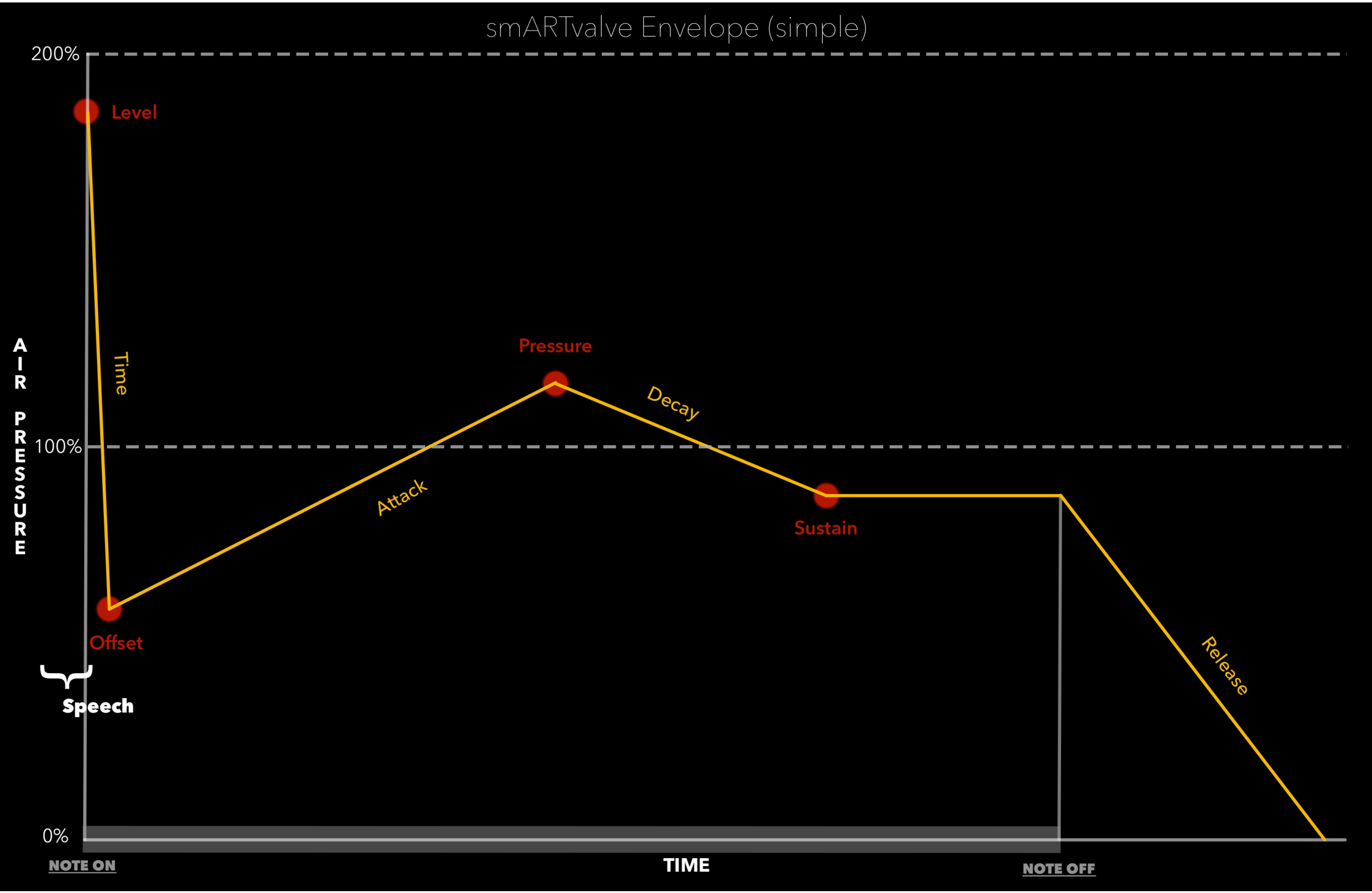
For a clear understanding of the parameters refer to smARTvalve Envelope Drawing on next page. Notice that not all parameters on the drawing are already functional within the smARtvalve's firmware. Only the ones listed on this list.

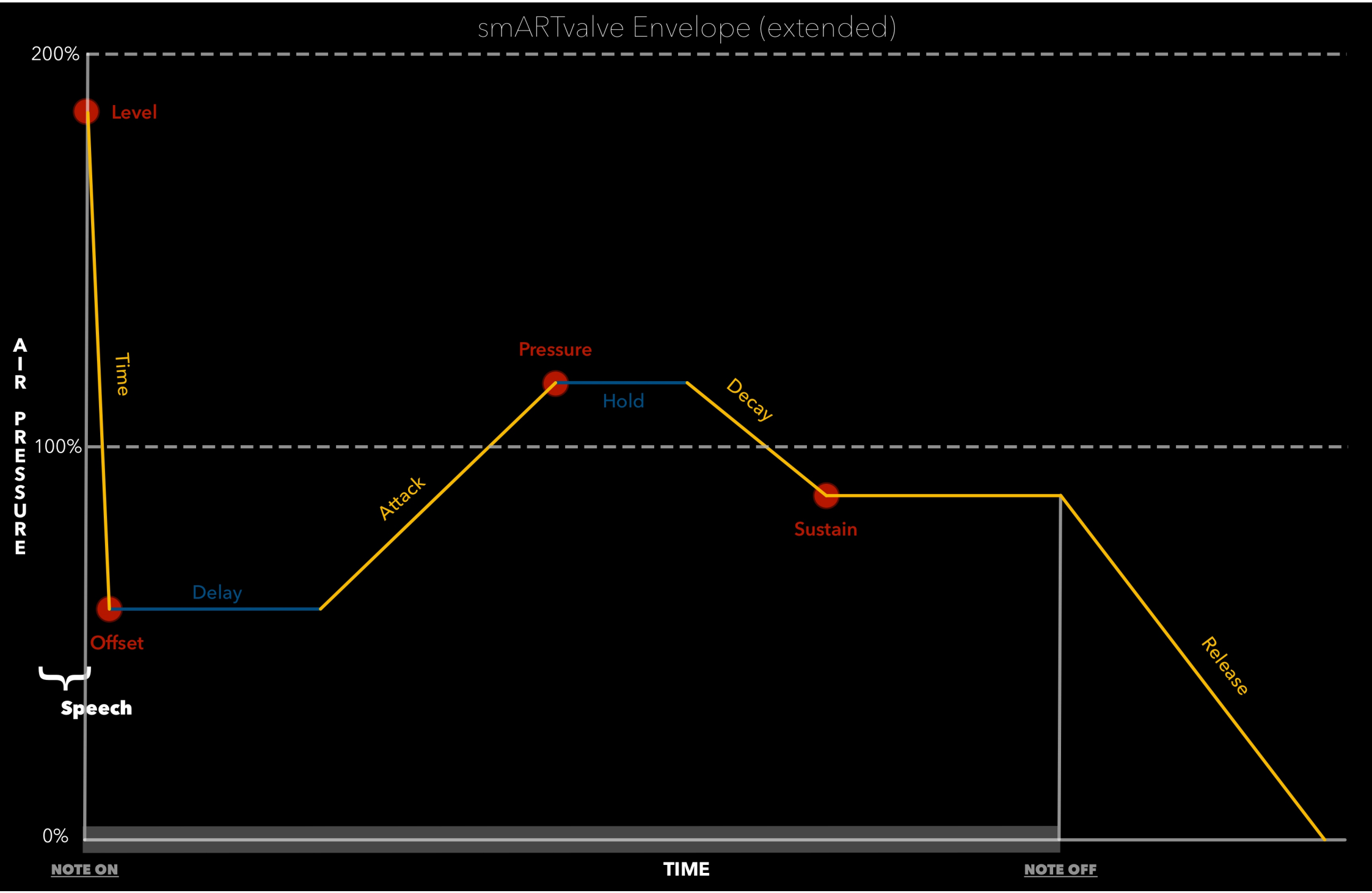
*Ascending CC Number*

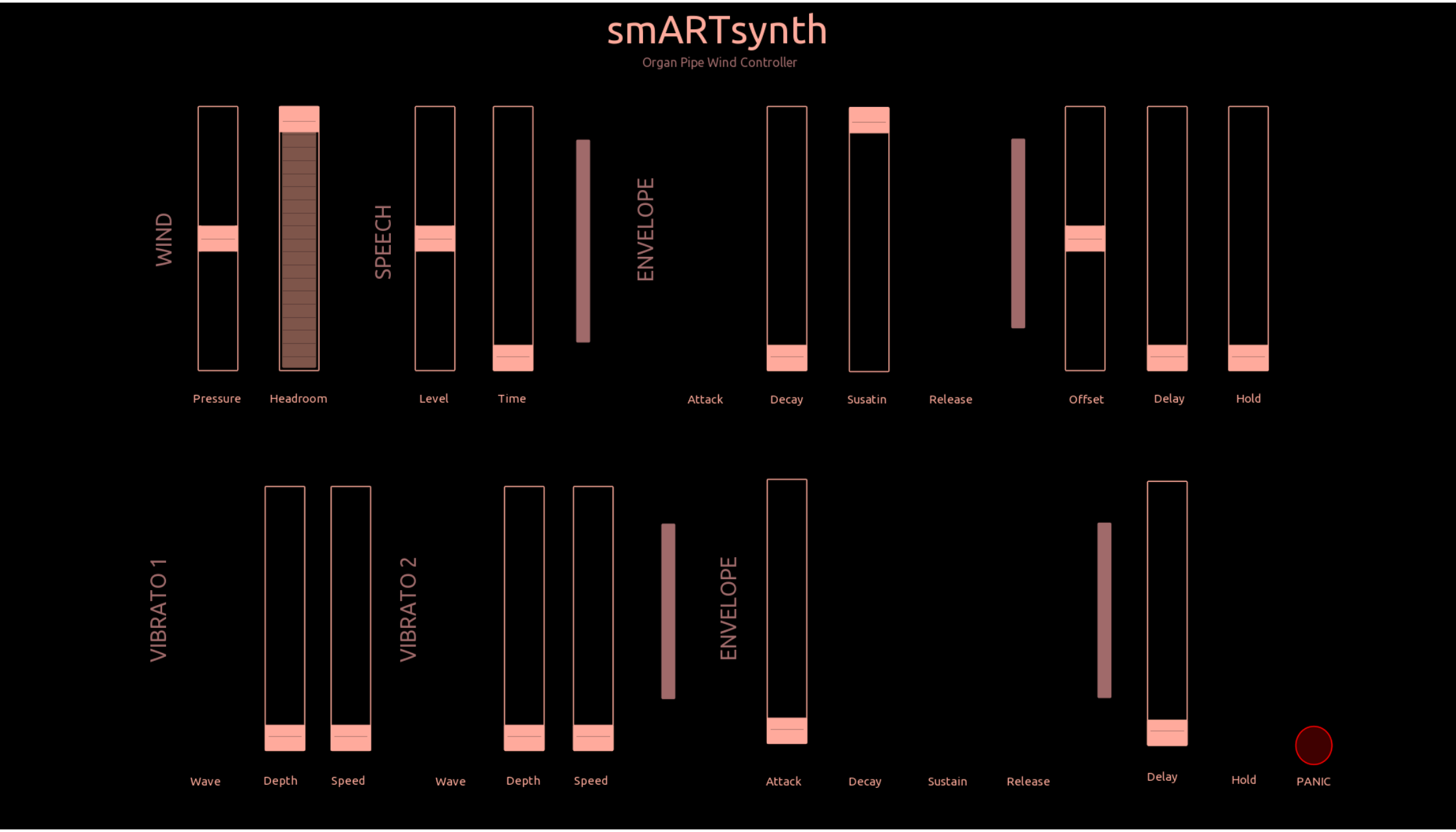
| Controller Name | Control |
|---|---|
| SWITCH ON | 38 |
| SWITCH OFF | 39 |
| Pressure | 40 |
| Headroom | 41 |
| Delay (*Vibratos*) | 42 |
| Attack (*Vibratos*) | 43 |
| Vibrato 1 Depth | 44 |
| Vibrato 1 Speed | 45 |
| Vibrato 2 Depth | 46 |
| Vibrato 2 Speed | 47 |
| Offset | 48 |
| Delay (*Wind Pressure*) | 49 |
| Hold (*Wind Pressure*) | 50 |
| Decay (*Wind Pressure*) | 51 |
| Sustain (*Wind Pressure*) | 52 |
| Sustain (Pedal) | 64 |
| Level (Speech) | 72 |
| Time (Speech) | 73 |
| Full Reset | 121 |

*In Groups*

| Controller Name | Control |
|---|---|
| Sustain (Pedal) | 64 |
| Full Reset | 121 |
| WIND PRESSURE | |
| Pressure | 40 |
| Headroom | 41 |
| SPEECH | |
| Level (Speech) | 72 |
| Time (Speech) | 73 |
| WIND ENVELOPE | |
| Offset | 48 |
| Delay (Wind Pressure) | 49 |
| Hold (Wind Pressure) | 50 |
| Decay (Wind Pressure) | 51 |
| Sustain (Wind Pressure) | 52 |
| VIBRATO | |
| Vibrato 1 Depth | 44 |
| Vibrato 1 Speed | 45 |
| Vibrato 2 Depth | 46 |
| Vibrato 2 Speed | 47 |
| VIBRATO ENVELOPE | |
| Delay (Vibratos) | 42 |
| Attack (Vibratos) | 43 |
| SWITCHES | |
| SWITCH ON | 38 |
| SWITCH OFF | 39 |


smARTvalve Envelope (simple)
200%
100%
0%
AIR PRESSURE
Level
Time
Pressure
Decay
Attack
Sustain
Offset
Speech
Release
NOTE ON
TIME
NOTE OFF


smARTvalve Envelope (extended)
200%
100%
0%
AIR PRESSURE
Level
Time
Offset
Speech
Delay
Attack
Pressure
Hold
Decay
Sustain
Release
NOTE ON
TIME
NOTE OFF

# smARTsynth

Organ Pipe Wind Controller